

Algorithms Problem Solving

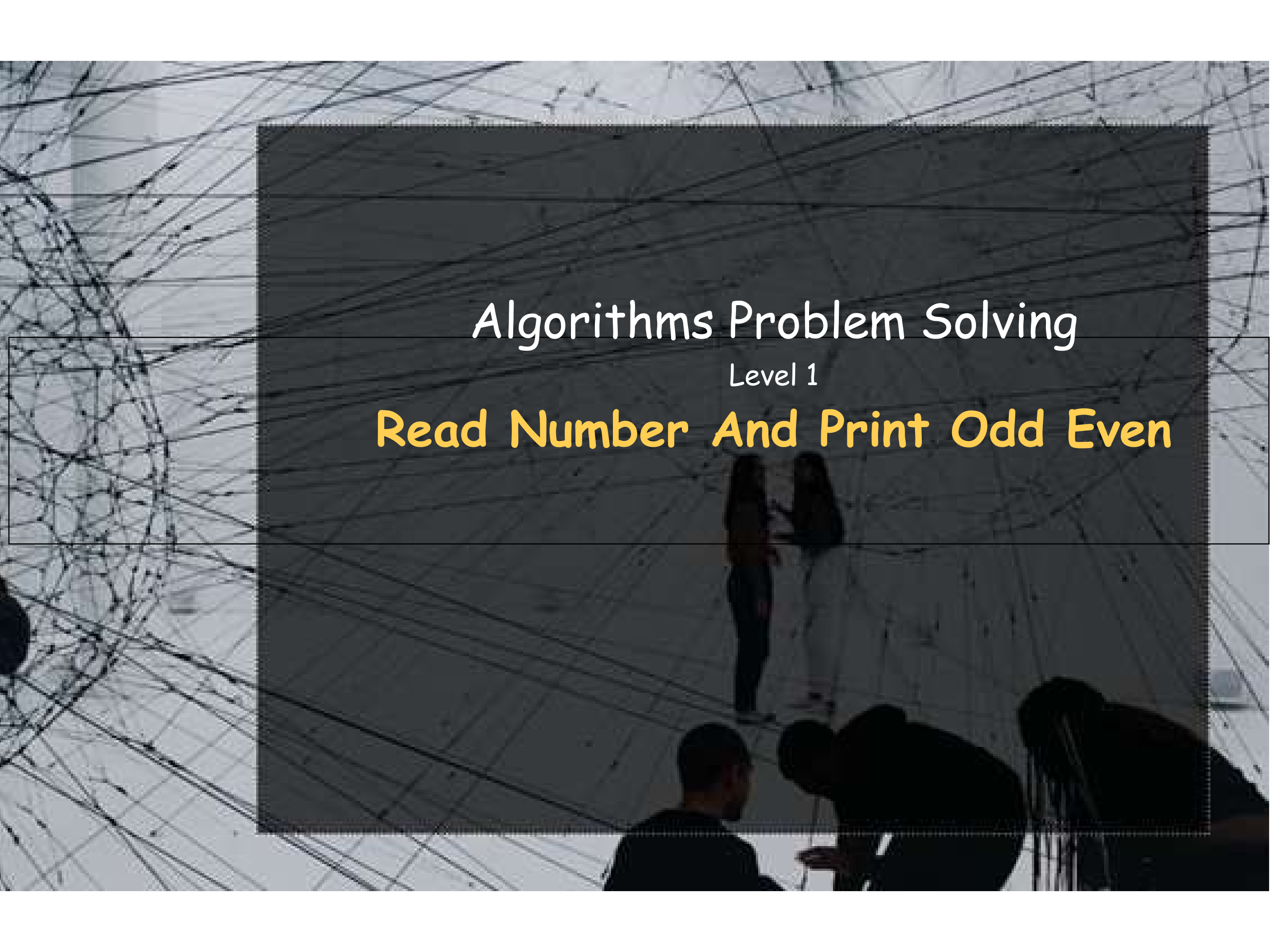
Level 1

Read Number And Print
Odd Even

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Read Number And Print Odd Even



Definition

- ◆ This program reads a number from the user, determines whether it is odd or even, and prints the result.
- ◆ It uses an enum and three separate functions to divide the program into clear responsibilities.

Explanation

- ◆ The program uses the enum `enNumberType` to represent the two possible results:
 `ODD = 1`
 `EVEN = 2`
- ◆ The program has three functions:
 - `readNumber()` reads and returns the user's number.
 - `checkNumberType()` checks the number using the modulo operator `%`. If `num % 2 == 0`, ◆ the number is even; otherwise, it is odd.
- ◆ `printResult()` receives the enum value and prints the appropriate message.
- ◆ In `main()`, the functions are combined:
- ◆ `printResult(checkNumberType(readNumber()));`
- ◆ The functions execute from the inside outward: `readNumber()` → `checkNumberType()` → `printResult()`.

Steps in main()

1. Read a number from the user.
2. Pass the number to checkNumberType().
3. Use % 2 to determine whether the number is even or odd.
4. Return EVEN or ODD.
5. Pass the result to printResult().
6. Print whether the number is even or odd.

💡 Example

If the user enters:

7

$$7 \% 2 = 1$$

So checkNumberType() returns ODD.

The program prints:

The number is odd.

If the user enters 10:

$$10 \% 2 = 0$$

So checkNumberType() returns EVEN.

➡ Result

The program correctly identifies the entered number as either odd or even.

Note

- ◆ The modulo operator % returns the remainder after integer division.



Comparison

EVEN

- ✓ The number is divisible by 2.
- ✓ $\text{num} \% 2 == 0$.
- ✓ Example: 2, 4, 10.

ODD

- ✓ The number is not divisible by 2.
- ✓ $\text{num} \% 2 != 0$.
- ✓ Example: 1, 3, 7.

✦✦ Summary

- ✓ The program uses functions to separate reading, checking, and printing.
- ✓ The modulo operator % is used to determine whether a number is odd or even.
- ✓ An enum is used to represent the ODD and EVEN results.

Thank you 😊